Does Air Quality Explain Adult Asthma Across U.S. Counties?

TruBridge Health Analytics Externship | Data-Driven Story

Research Question

Is county-level air quality (median AQI) associated with adult asthma prevalence across U.S. counties, and what does the pattern suggest for the Los Angeles area?

I started from a broader question (how air pollution relates to asthma in urban communities) and considered several narrower ones before choosing this one:

- Which pollutants (PM2.5, ozone, NO2) are most associated with higher asthma prevalence?
- Do counties with more unhealthy air-quality days have higher asthma prevalence?
- How much of the variation in county asthma is explained by air quality versus county size and state?

I selected the question above because both datasets can answer it directly at the county level. My original question also asked about income and race/ethnicity, but neither dataset includes them, so county population and state are the only controls used here. Adding income data is a next step.

Step 1: Context

Asthma affects millions of U.S. adults, and air pollution is one of the most widely studied triggers. The Los Angeles area is a natural place to ask this question: it has long faced air-quality challenges, and the Los Angeles County public health research I reviewed shows that asthma outcomes vary across neighborhoods and with income and race.

The analysis uses two public datasets:

- **CDC PLACES, County Data (2025 release):** modeled estimates of current asthma prevalence among adults aged 18 and older for 3,143 U.S. counties.
- **EPA Annual AQI by County (2026):** median AQI, unhealthy-day counts, and the number of days each pollutant drove the AQI, for 827 counties with air monitors.

After cleaning and merging on state and county name, **769 counties** had both an asthma estimate and an AQI value. Because only monitored counties are included, the sample leans toward larger counties (median population about 120,500, compared with about 26,000 for all U.S. counties).

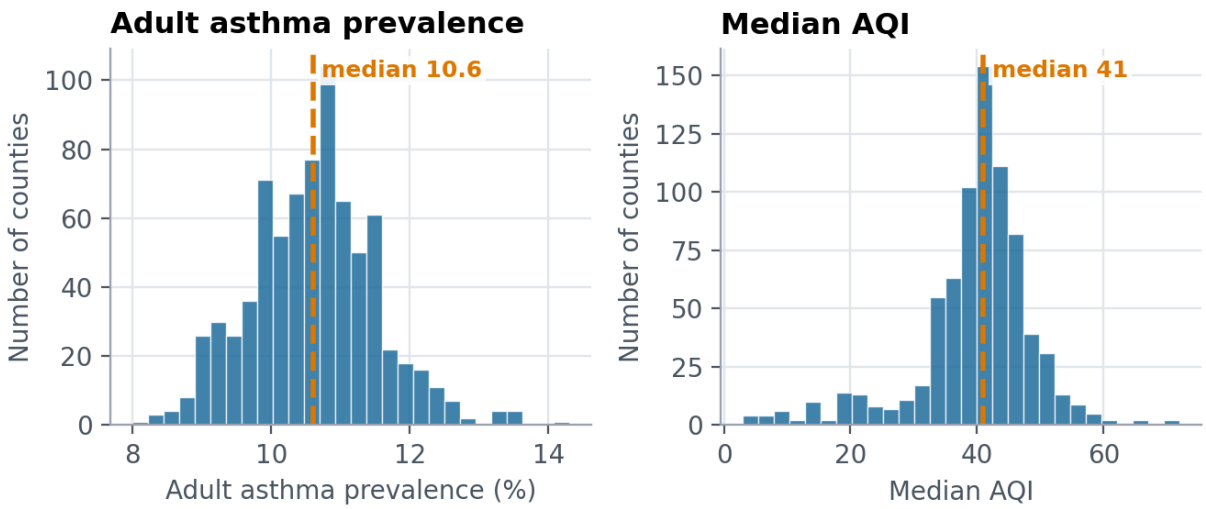
Step 2: Key Findings

Finding 1: Counties look alike on both measures

Adult asthma prevalence is roughly bell-shaped, averaging 10.6% with a range of 8.0% to 14.3%. Median AQI centers near 40, with a tail of cleaner counties. Few counties stand out on either measure, so large differences between counties are unlikely to appear.

Figure 1. Distribution of adult asthma prevalence and median AQI across 769 counties.

Most counties sit near 10.6% asthma and a median AQI of 41 (769 counties)



Finding 2: Worse air does not mean more asthma

The correlation between median AQI and asthma prevalence is essentially zero (r = -0.04). Unhealthy-day share (r = -0.11), PM2.5 days (r = 0.00), and ozone days (r = -0.03) tell the same story. Counties in the lowest and highest AQI quartiles averaged 10.67% and 10.61% asthma prevalence. Five methods agree that there is no meaningful difference: Welch's t-test (p = 0.48), Mann-Whitney U (p = 0.82), ANOVA across dominant-pollutant groups (p = 0.35), a chi-square test on above-median asthma (p = 0.37), and a regression controlling for population (AQI coefficient 0.005, p = 0.19).

Figure 2. Adult asthma prevalence vs. median AQI for each county.

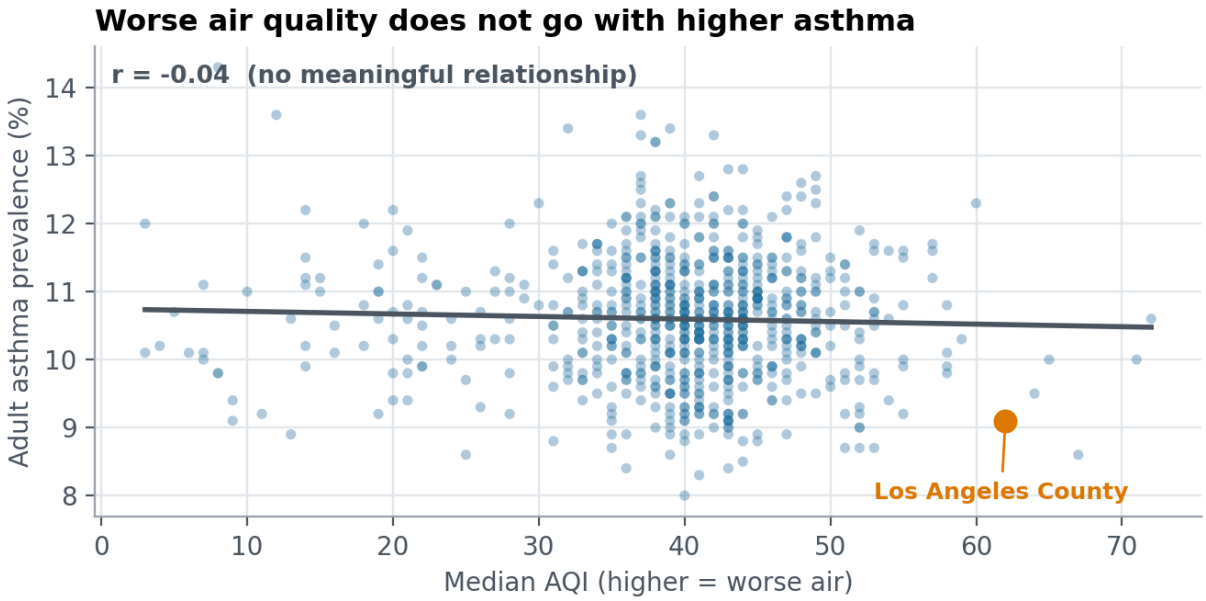
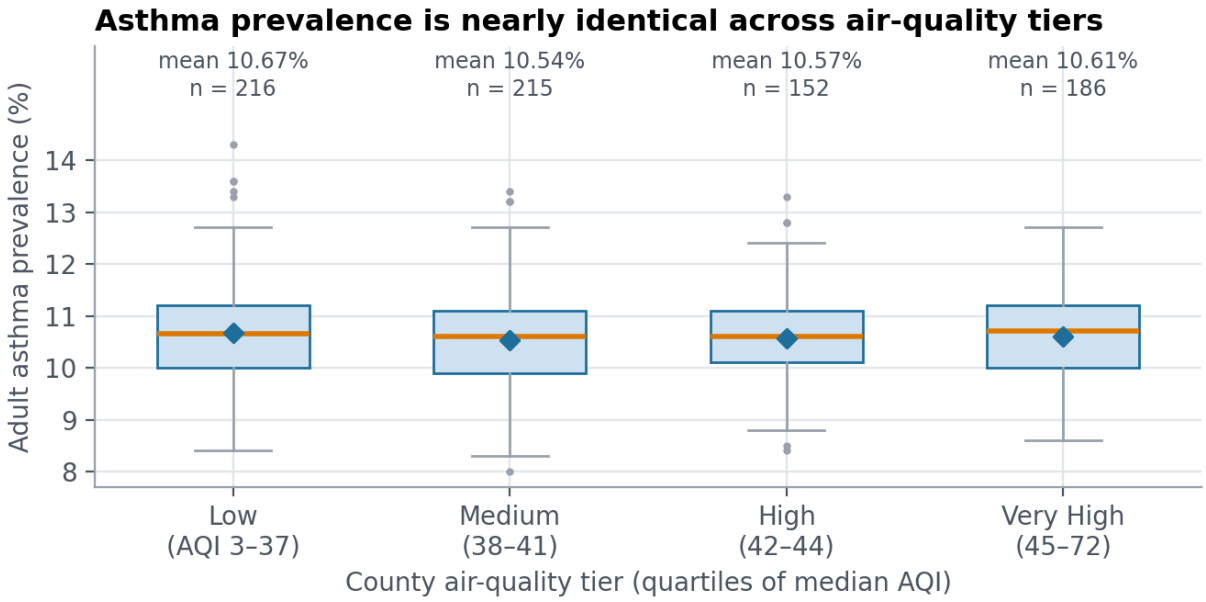


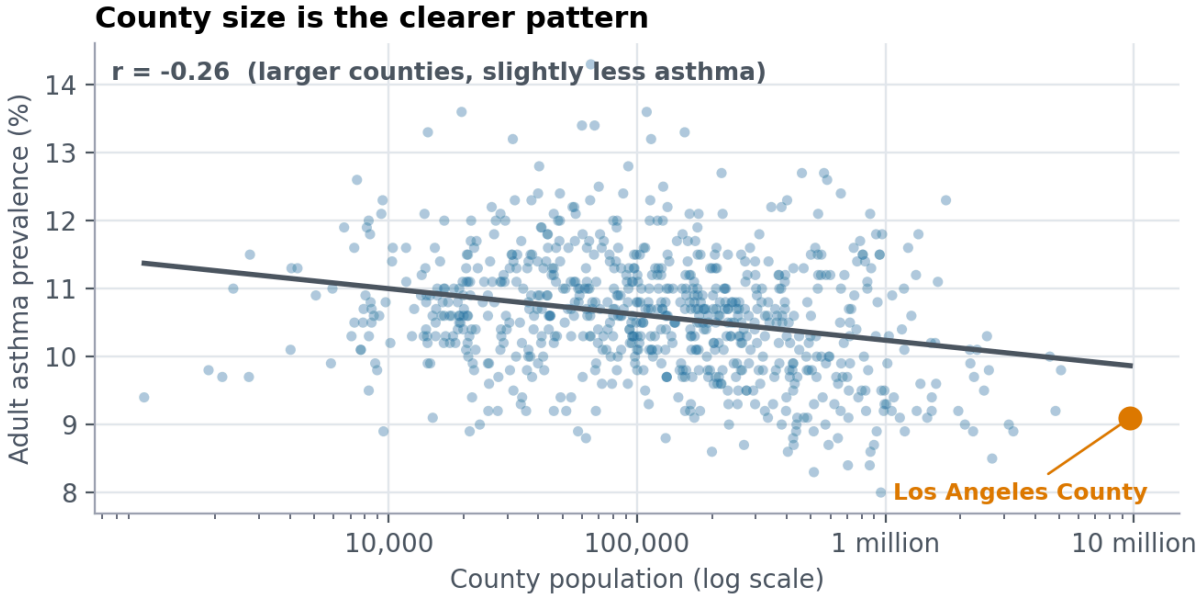
Figure 3. Adult asthma prevalence by AQI quartile.



Finding 3: County size and state matter more than air quality

Larger counties report slightly lower asthma prevalence (r = -0.26), and population is the only strong predictor in the regression ($p < 0.001$). Adding state effects raises the share of variation explained from 9% to 67%, yet AQI remains non-significant ($p = 0.48$). State differences explain most of the variation in county asthma, which partly reflects how PLACES models small areas from state-level survey data.

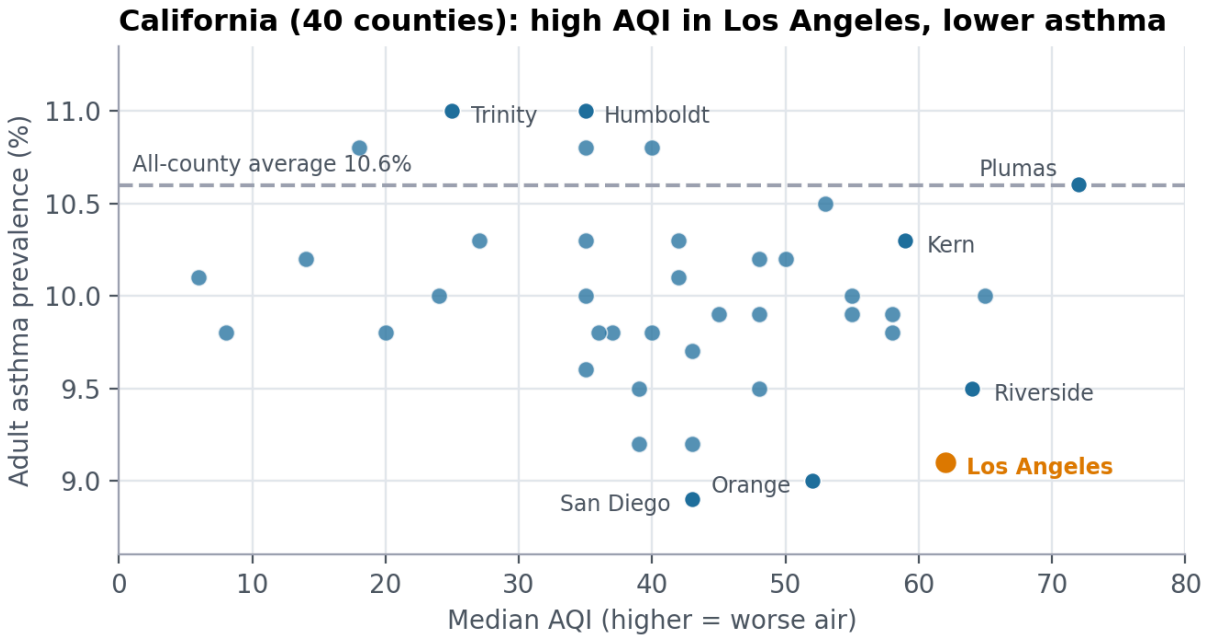
Figure 4. Adult asthma prevalence vs. county population (log scale).



Finding 4: The Los Angeles area

Among the 40 California counties with data, Los Angeles County has one of the highest AQI values in the state (62) but below-average asthma (9.1%, compared with a sample average of 10.6%). Orange (9.0%) and San Diego (8.9%) look similar, while the highest asthma appears in northern California counties such as Trinity and Humboldt (11.0%) with moderate AQI. County averages hide neighborhood differences, such as exposure near freeways and ports reported in Los Angeles research, so this does not mean local air is harmless.

Figure 5. California counties: adult asthma prevalence vs. median AQI.



Step 3: Key Takeaways and Next Steps

Takeaways

- In this data, counties with worse air quality do not have higher adult asthma prevalence, and five different methods agree.
- A county's state and size say more about its asthma level than its AQI does.
- This is a limit of the data, not proof that air pollution is harmless: county averages, adult-only estimates, and missing socioeconomic data can hide real effects.

Recommended next steps

- Move to census-tract data in Los Angeles County to capture neighborhood-level exposure.
- Add income, poverty, and race/ethnicity from Census ACS data to test for confounding.
- Study children, who may be more sensitive to air pollution, using child asthma data.
- Use multiple years of AQI data to match the survey period and look at trends.

Limitations

- PLACES asthma values are modeled estimates and cover adults aged 18 and older only.
- Only monitored counties (827) have AQI data, which skews the sample toward larger counties. In addition, 187 counties lack an asthma estimate, 27 AQI counties could not be matched by name, and the 2026 AQI file was produced mid-year, with many counties showing fewer than 100 monitored days.
- Three counties (Baltimore, Fairfax, and Roanoke) matched both a city and a county row. Removing them does not change the results.
- Both files are single-year snapshots, so no time trends could be analyzed.
- Correlation does not show causation.